Diagnostic Logs: Model 3 Battery Cases (SYNTHETIC TEST DOCUMENT)

> **SYNTHETIC TEST DATA.** These logs were invented to test the FixSkrew pipeline. They are not from any real vehicle. Every vehicle, VIN, reading and date is fictional.

Document ID: SYN-M3-LOG-001 · Language: English

Reading guide: voltages in volts (V), temperatures in degrees Celsius (°C), current in amperes (A), insulation in ohms per volt (Ω/V), state of charge (SOC) and state of health (SOH) in percent (%). "Group" means one series group of parallel cells; this test pack has 96 groups in 4 modules.

Log A: Reduced range, balanced pack, no fault

Vehicle: Model 3 Long Range (synthetic), 2021, 38,200 km, VIN 5YJ3E1EA7MF000001 (fictional) **Date:** 2025-01-14 · **Complaint:** "Range dropped by about 15 % this winter."

A.1 Trouble codes

No active or stored codes.

A.2 Pack readings at rest (vehicle asleep for 3 hours)

Measurement	Value	Unit
State of charge	62	%
State of health	94	%
Pack voltage	369.8	V
Highest cell group voltage	3.852 (group 41, module 2)	V
Lowest cell group voltage	3.838 (group 17, module 1)	V
Cell voltage spread	0.014	V
Pack temperature (average)	4.5	°C
Module temperature spread	1.2	°C
Insulation resistance	1,850	Ω/V
Idle current	0.2	A

A.3 Findings

Cell spread is far below the investigation threshold, state of health is above 90 %, insulation is healthy and there are no codes. The pack temperature of 4.5 °C explains most of the reduced range: cold cells deliver less usable energy and cabin heating draws power. No battery fault is indicated by this log.

A.4 Recommended

Explain winter range loss, advise preconditioning on the charger, check tyre pressure.

Log B: Cell voltage imbalance (weak group in module 3)

Vehicle: Model 3 Standard Range Plus (synthetic), 2020, 96,400 km, VIN 5YJ3E1EB1LF000002 (fictional) **Date:** 2025-02-03 · **Complaint:** "Range is down, and charging seems to stop early at the same point."

B.1 Trouble codes

Code	Status	Description
P0B3C	Active	Cell voltage imbalance above limit
P0A7F	Stored	Battery pack deterioration

B.2 Pack readings at rest (after 3 hours of rest, SOC 71 %)

Measurement	Value	Unit
State of health	83	%
Pack voltage	375.2	V
Highest cell group voltage	3.911 (group 62, module 3)	V
Lowest cell group voltage	3.771 (group 63, module 3)	V
Cell voltage spread	0.140	V
Pack temperature (average)	21.0	°C
Module temperature spread	2.0	°C
Insulation resistance	1,640	Ω/V

B.3 Balancing charge result

After a balancing charge to 100 % held for 4 hours and 2 hours of rest, the spread was re-measured at 0.126 V. Group 63 of module 3 was again the lowest.

B.4 Load test (drive, 150 A discharge)

Measurement	Value	Unit
Lowest cell group voltage under load	3.402 (group 63)	V
Neighbouring groups (typical)	3.590	V
Drop of group 63 compared with neighbours	0.188	V
Cell voltage spread under load	0.206	V

B.5 Findings

- Spread at rest is 0.140 V, above the 0.080 V action limit, and did not recover after balancing. This rules out simple imbalance from partial charging.
- The same group (63, module 3) is lowest at rest and under load, and drops by 0.188 V more than its neighbours under 150 A. This points to high internal resistance in that group.
- Pack temperature is normal and insulation is healthy, so neither thermal nor isolation problems explain it.
- Sense connector check was not recorded in this log.

B.6 Recommended

Reseat and inspect the module 3 sense connector to exclude a false reading, then re-test. If group 63 remains weak, replace module 3. Verify with a balancing charge and a load test.

Log C: Isolation fault after wet weather

Vehicle: Model 3 Long Range (synthetic), 2021, 61,000 km, VIN 5YJ3E1EA3MF000003 (fictional) **Date:** 2025-03-22 · **Complaint:** "Red HV warning after driving through heavy rain. Car would not drive. It worked again next day."

C.1 Trouble codes

Code	Status	Description
P0AA6	Stored (3 occurrences)	High-voltage isolation fault

C.2 Readings

Measurement	Value	Unit
Insulation resistance, pack to chassis	410	Ω/V
Insulation resistance after 2 hours in a heated bay	520	Ω/V
Coolant level	low, 0.4 l below the maximum mark	
Pack temperature (average)	19.0	°C
Cell voltage spread	0.011	V
State of health	95	%

C.3 Branch test (each load disconnected in turn)

Branch disconnected	Insulation after disconnection
Front drive unit	415 Ω/V (no change)
Rear drive unit	418 Ω/V (no change)
Cabin heater	420 Ω/V (no change)
Air-conditioning compressor	422 Ω/V (no change)
Pack alone	425 Ω/V (still low)

C.4 Findings

- The loads are not the cause. The low insulation remains with the pack isolated, so the fault is inside the pack or at its connections.
- The coolant level is low without an external leak, and the fault appears after rain. This pattern matches coolant or water inside the enclosure, and also matches the symptoms of campaign SYN-RC-24-002 (cooling plate coolant leak) for a 2021 long-range vehicle.
- Cell voltage spread and state of health are normal, so the cells themselves look healthy.

C.5 Recommended

Check whether campaign SYN-RC-24-002 has been completed. Pressure-test the cooling loop. If a leak into the enclosure is confirmed, the pack must be removed, dried and inspected by a trained specialist. Do not return the vehicle to service with insulation below 100 Ω/V. Re-measure after repair.

Log D: Slow DC charging in cold weather

Vehicle: Model 3 Standard Range Plus (synthetic), 2022, 24,900 km, VIN 5YJ3E1EB5NF000004 (fictional) **Date:** 2025-01-29 · **Complaint:** "Fast charging takes twice as long in winter."

D.1 Trouble codes

None.

D.2 Charge session log (DC fast charger, ambient −6 °C)

Time into session (min)	Pack temp (°C)	SOC (%)	Charge power (kW)
0	−2.0	18	22
5	1.5	22	38
10	6.0	29	62
15	11.0	38	88
20	16.5	48	104
30	24.0	66	98

D.3 Preconditioning test

After 10 minutes of preconditioning the pack warmed from −2.0 °C to 7.5 °C. Pump and heater commands and feedback matched.

D.4 Findings

The charge rate recovers as the pack warms, the heater works, and there are no codes. This is normal cold-weather behaviour covered by SYN-TSB-24-014. No fault.

D.5 Recommended

Advise navigating to the charger with preconditioning enabled.

Log E: Interlock open, intermittent loss of drive

Vehicle: Model 3 Long Range (synthetic), 2019, 118,000 km, VIN 5YJ3E1EA9KF000005 (fictional) **Date:** 2025-04-08 · **Complaint:** "Lost drive power twice while driving over rough road. Restarted and it was fine."

E.1 Trouble codes

Code	Status	Description
P0A0D	Stored (4 occurrences)	HV system interlock circuit open

E.2 Freeze frame (last occurrence)

Item	Value	Unit
Vehicle speed	52	km/h
Pack current at the time	85	A
Pack voltage	351.0	V
Pack temperature	24.0	°C
Cell voltage spread	0.019	V
Interlock loop resistance (when closed)	normal	

E.3 Findings

- The pack is healthy: normal spread and voltage, and the fault happens only over rough road.
- An intermittent open interlock under vibration is the typical effect of a poorly seated HV connector.
- The vehicle is a 2019 model, inside the population of campaign SYN-RC-24-001 (connector latch). The campaign sticker was not found on the junction box cover.

E.4 Recommended

Apply campaign SYN-RC-24-001. Inspect junction box connector pins and latches. Clear the code and test over rough road. No battery replacement is indicated.

Log F: Pack deterioration with state-of-health report

Vehicle: Model 3 Standard Range Plus (synthetic), 2019, 187,000 km, VIN 5YJ3E1EB0KF000006 (fictional) **Date:** 2025-05-06 · **Complaint:** "Range keeps shrinking. Now only 220 km on a full charge."

F.1 Trouble codes

Code	Status	Description
P0A7F	Active	Battery pack deterioration

F.2 Readings (after software update SYN-TSB-24-018 and a relearn cycle)

Measurement	Value	Unit
State of health (after update)	68	%
Measured usable capacity (capacity test)	36.8	kWh
Cell voltage spread at rest	0.021	V
Cell voltage spread under load	0.088	V
Pack temperature (average)	22.0	°C
Module temperature spread	1.4	°C
Insulation resistance	1,520	Ω/V

F.3 Findings

- Spread is small at rest and under load, so the loss is not caused by a single weak group.
- State of health stayed low after the software update and relearn, and the measured capacity agrees with the reported state of health. The value is no longer attributable to an estimation error.
- The mileage and age are consistent with uniform ageing of all cells.

F.4 Recommended

Explain that this is normal end-of-life capacity loss, not a defect. Discuss options: continue use with reduced range, or replace the pack or its modules. Document the capacity test.

Log G: Pre-charge timeout and contactor fault

Vehicle: Model 3 Long Range (synthetic), 2019, 134,000 km, VIN 5YJ3E1EA4KF000007 (fictional) **Date:** 2025-06-17 · **Complaint:** "Sometimes the car takes a long time to become ready, and then a loud clunk."

G.1 Trouble codes

Code	Status	Description
P0AE2	Active	Pre-charge contactor circuit fault

G.2 Readings

Measurement	Value	Unit
12 V battery voltage	12.6	V
Pre-charge time to 95 % bus voltage	3.8	s
Pre-charge resistor	open circuit	
Cell voltage spread	0.010	V
State of health	91	%

G.3 Findings

12 V supply is healthy, which rules out a false contactor fault from a weak 12 V battery. The pre-charge resistor measures open circuit and the pre-charge time of 3.8 s is above the 3 s limit. The cells and pack are healthy.

G.4 Recommended

Replace the HV junction box assembly (includes the pre-charge resistor and contactors). Verify pre-charge time below 1.5 s after repair.

Log H: Charge port overheating complaint

Vehicle: Model 3 Long Range (synthetic), 2020, 72,000 km, VIN 5YJ3E1EA6LF000008 (fictional) **Date:** 2025-07-02 · **Complaint:** "Charging stops after a few minutes with a charge port message."

H.1 Trouble codes

Code	Status	Description
P1A00	Stored	Charge port temperature too high

H.2 Readings during a test charge (AC, 11 kW)

Time (min)	Charge port temp reported (°C)	Surface temp measured with IR (°C)
0	24	24
3	88	26
6	88	27

H.3 Findings

The reported charge port temperature jumps to 88 °C within 3 minutes while the surface temperature of the port is only 26 °C. The reported value is not credible, which points to the temperature sensor circuit or wiring. This is the condition described in recall SYN-RC-23-007 (charge port temperature sensor wiring).

H.4 Recommended

Check campaign SYN-RC-23-007. Replace the sensor harness and connector. Do not replace the charge port before testing the sensor circuit.

Log I: Several faults together (do not conclude from a single reading)

Vehicle: Model 3 Standard Range Plus (synthetic), 2021, 49,000 km, VIN 5YJ3E1EB8MF000009 (fictional) **Date:** 2025-08-19 · **Complaint:** "Car limited power, then a warning, now charging is slow."

I.1 Trouble codes

Code	Status
P0A81	Active
P0A9C	Active
P0B3C	Stored

I.2 Readings

Measurement	Value	Unit
Module 1 temperature	31.0	°C
Module 2 temperature	30.5	°C
Module 3 temperature	30.8	°C
Module 4 temperature	58.0	°C
Pump command	100	%
Pump actual speed	0	%
Cell voltage spread at rest	0.071	V
State of health	92	%

I.3 Findings

- Three of the four modules agree at about 31 °C. Module 4 reports 58 °C. A reading this far from three neighbours may be a sensor fault (P0A9C) or a real hot spot. A single reading cannot decide. Measure with a contact thermometer.
- The pump is commanded to 100 % but does not turn: a pump or its circuit fault (P0A81), which would reduce cooling and explain the power limits and slow charging.
- The cell spread of 0.071 V is in the investigation band and may be a consequence of the temperature difference rather than a weak cell. Re-measure after cooling.

I.4 Recommended

Prioritise the pump fault. Check pump fuse, connector and 12 V supply. Verify module 4 with an independent thermometer. Re-measure the cell spread after the thermal issue is repaired before concluding that a module is bad.